

Born-Haber cycle (HL)

Higher level (HL)

Learning outcomes

By the end of this section you should be able to:

- Interpret and determine values from a Born-Haber cycle for compounds composed of univalent and divalent ions.
- Understand the following terms and use those values in Born-Haber cycle calculations: ionisation energies, enthalpy of atomisation (using sublimation and/or bond enthalpies), electron affinities, lattice enthalpy, enthalpy of formation.

Salts are common compounds found in our bodies, in the environment, and in the laboratory. But how are they formed from their elements? This is not a common reaction in the environment, but under certain conditions, can be completed in a laboratory. The elements would need to undergo a number of transformations in order to be in a cationic or anionic form ready for bonding in a crystal lattice. The metal would begin as a solid that would need to be separated from the other atoms held by metallic bonding and the non-metal would need to be separated from its covalent bonds and intermolecular forces. And once separated, the metal would need to lose electrons with the non-metal gaining electrons. How many steps are involved and what is the sequence?



Figure 1. Great Salt Lake, Utah, USA.

Credit: Alex Potemkin, Getty Images

Making connections

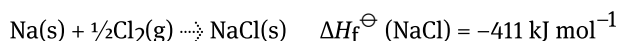
Section S2.1.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/>) showed us that the ionic bond is formed by electrostatic attractions between oppositely charged ions and in section S2.1.3a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-ionic-compounds-id-43558/>) we saw that ionic compounds exist as three-dimensional lattice structures, represented by empirical formulas. How can we assess the magnitude of lattice enthalpy from the chemical structure?

Processes to transform atoms and molecules into salts

As we learned in sections S2.1.2 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/ionic-bonding-id-43557/>) and S2.1.3a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-ionic-compounds-id-43558/>) three-dimensional lattice structures, or salts are composed of ions which are bonded ionically. Their formation can be expressed with the enthalpy of formation, $\Delta H_f^\ominus$ defined as the enthalpy change that would occur when one mole of a compound is formed from its elements in their standard states under standard conditions.

But it is not as simple as mixing a metal with a non-metal – many steps are required.

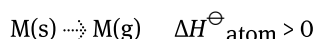
To illustrate these steps, let's investigate the formation of NaCl from its elements, Na(s) and Cl₂(g). This overall reaction would be measured with the standard enthalpy of formation:



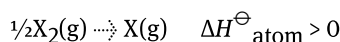
Firstly, solid sodium metal would need to be transformed into gaseous sodium which would be calculated with the enthalpy of atomisation.

- Enthalpy of atomisation, $\Delta H_{\text{atom}}^\ominus$: the enthalpy change when one mole of gaseous atoms is formed from an element in its standard state.

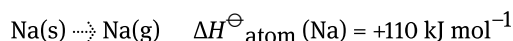
For metals:



For diatomic species:



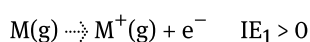
Therefore:



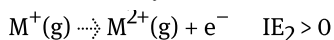
Once the metal is in the gaseous state, electrons need to be removed to create cations. This is measured using the ionisation energy:

- Ionisation energy, IE: the energy required to remove one mole of electrons from one mole of gaseous atoms. For some metals with multiple valence electrons, the second ionisation energy may be required.

IE₁:



IE₂:



Therefore:



Next, we need to think about how to change the non-metal, $Cl_2(g)$ into anions. The first step is breaking the bond between the chlorine atoms. This is measured using bond energy:

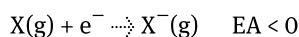
- Bond energy, E: the energy required to break one mole of bonds in the gaseous state.

For chlorine, we can write this as follows:



With the non-metal now existing as an atom instead of a molecule, electrons need to be added to form an anion. This is measured with electron affinity:

- Electron affinity, EA: the energy released when one mole of electrons is added to one mole of gaseous atoms.



Therefore:



Study skills

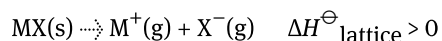
Finding the data required when solving questions with lattice enthalpy can be one of the trickiest parts! Here is where to find the data:

- Enthalpy of atomisation, $\Delta H^\ominus_{\text{atom}}$: will be given in the question or you will be asked to solve for it.
- Ionisation energy, IE : is found in section 9 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/>) of the DP chemistry data booklet.
- Bond energy, E : found in section 12 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/bond-enthalpies-or-average-bond-enthalpies-at-id-45114/>) of the DP chemistry data booklet.
- Electron affinity, EA : found in section 9 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/first-ionisation-energy-electron-affinity-and-id-45111/>) of the DP chemistry data booklet .
- Standard enthalpy of formation, $\Delta H^\ominus_f$: found in section 13 (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/thermodynamic-data-selected-compounds-id-45115/>) of the DP chemistry data booklet .

Born-Haber cycle

Recall from section S2.1.3a (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/structure-of-ionic-compounds-id-43558/>) that lattice enthalpy, $\Delta H^\ominus_{\text{lattice}}$, gives us information about the strength of ionic bonds in an ionic compound. Lattice enthalpy is the process by which gaseous ions are formed from

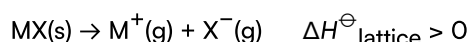
one mole of solid ionic compound. This process is endothermic and can be represented by the following equation where MX(s) represents one mole of solid ionic compound.



Hess's law provides us with a tool to determine the lattice enthalpy of ionic compounds. This is particularly powerful since it is not possible to determine the lattice enthalpy of an ionic compound directly by experiment.

Study skills

Pay careful attention to the sign of lattice enthalpy. The IB defines lattice enthalpy as the endothermic process of the crystal lattice breaking into its ion:



Values for experimental lattice enthalpies at 298.15 K are found in [section 16](https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/lattice-enthalpies-at-29815-k-experimental-values-id-45118/) (<https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/lattice-enthalpies-at-29815-k-experimental-values-id-45118/>) of the DP chemistry data booklet.

The energy cycle that we use for determining the lattice enthalpy of an ionic compound is called a Born-Haber cycle. This energy cycle follows the law of conservation of energy (like all other energy cycles in this section) and it presents a visual representation of the process.

Nature of Science

Aspect: Models

The sign (+ or –) of lattice enthalpy varies, depending on the perspective of the definition: destruction or formation of the crystal lattice. Does it make more sense for the energy holding two ions together to be represented with a positive or negative sign?

Looking back to our previous example with NaCl, let's use the Born-Haber cycle to solve for its lattice enthalpy.

A Born-Haber cycle is structured as follows using the enthalpy of atomisation, ionisation energy, electron affinity, and enthalpy of formation as shown in **Figure 2**.

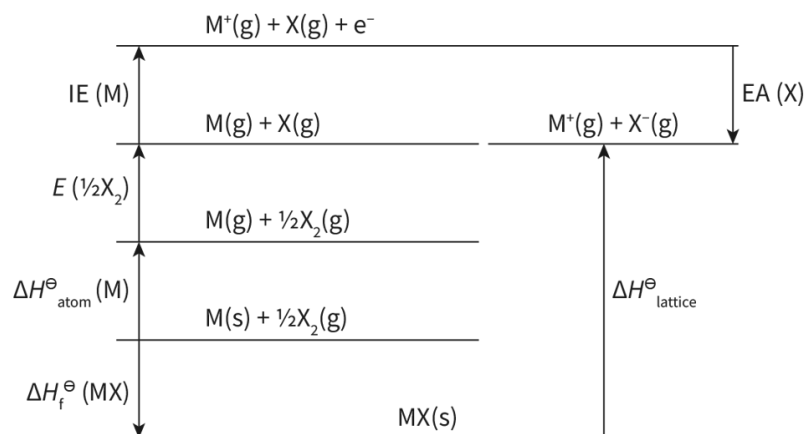


Figure 2. General Born-Haber cycle.

 More information for figure 2

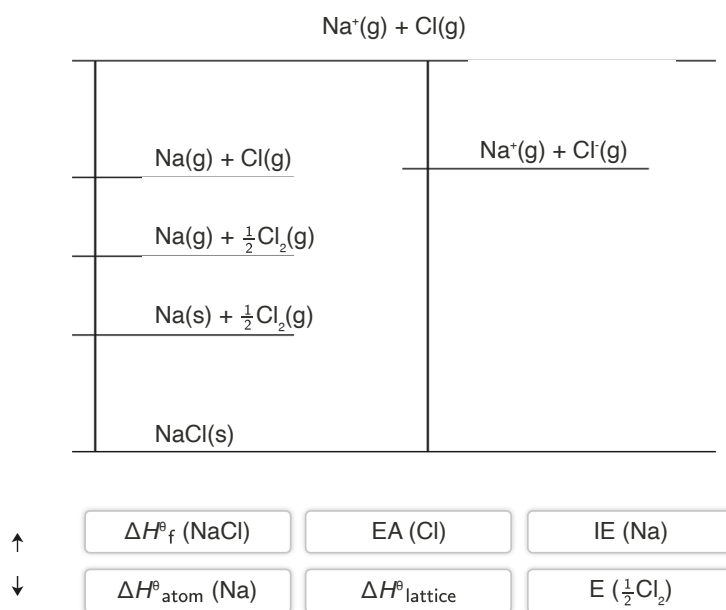
The image is a diagram illustrating a Born-Haber cycle, showing energy changes during the formation of an ionic compound from its elements. The diagram includes several horizontal lines connected by vertical arrows, each representing different energy stages:

1. At the bottom, the equation is $(MX(s))$, representing the solid state of the compound.
2. Above this, during the enthalpy of formation ($\Delta H_f^\ominus(MX)$), the conversion from solid to gas is represented as $(M(s) + \frac{1}{2}X_2(g))$.
3. The next stage is the enthalpy of atomization ($\Delta H_{\text{atom}}^\ominus(M)$), depicted as $(M(g) + \frac{1}{2}X_2(g))$.
4. This is followed by the bond dissociation energy ($E(\frac{1}{2}X_2)$), forming $(M(g) + X(g))$.
5. Ionization energy ($IE(M)$) is applied to obtain $(M^+(g) + X(g) + e^-)$.
6. Finally, electron affinity ($EA(X)$) leads to $(M^+(g) + X^-(g))$, completing the cycle.

Arrows indicate the direction and type of energy change at each stage, linking the equations sequentially.

[Generated by AI]

Try using **Interactive 1** to create the Born-Haber cycle for NaCl.



✓ Check

Interactive 1. Born-Haber Cycle for NaCl.

More information for interactive 1

An interactive diagram illustrates the Born-Haber cycle for NaCl. Users need to drag and drop the appropriate thermodynamic terms and directional arrows (upward and downward) to complete the cycle correctly.

The diagram is divided into two sections indicated with a top and bottom horizontal line, and two vertical lines: one on the left and one in the middle. The horizontal lines intersect with the vertical lines. The left section has three horizontal lines intersecting with the left vertical line. Similarly, the right section has one horizontal line intersecting with the middle vertical line. There is one drop box at each intersection, and therefore, there are eight drop boxes.

The diagram is titled $\text{Na}^+(\text{g}) + \text{Cl}(\text{g})$ mentioned at the top.

The left section features a vertical sequence of reactions with drop boxes arranged from top to bottom as follows:

$\text{Na}(\text{g}) + \text{Cl}(\text{g})$

Dropbox

$\text{Na}(\text{g}) + \frac{1}{2}\text{Cl}_2(\text{g})$

Dropbox

$\text{Na}(\text{s}) + \frac{1}{2}\text{Cl}_2(\text{g})$

Dropbox

$\text{NaCl}(\text{s})$

The right section features a vertical sequence of reactions and drop boxes arranged from top to bottom as follows:

Dropbox

$\text{Na}^+(\text{g}) + \text{Cl}^-(\text{g})$

Dropbox

Users need to place the correct upward or downward arrows and the appropriate thermodynamic terms in the drop boxes provided in the cycle. The drag-and-drop options provided at the bottom from left to right include

two rows of reads: Upward arrow, $\Delta H_f^\ominus(\text{NaCl})$, $\text{EA}(\text{Cl})$, $\text{IE}(\text{Na})$,

Downward arrow, $\Delta H_{\text{atom}}^\ominus(\text{Na})$, $\Delta H_{\text{lattice}}^\ominus$, and $\text{E}(\frac{1}{2}\text{Cl}_2)$.

This interactivity helps users to construct and interpret the Born-Haber cycle for the formation of NaCl, demonstrating the enthalpy changes involved in the process.

Read below for the solution.

The arrows on the leftmost intersections from top to bottom in the table include upward arrow, downward arrow, upward arrow, downward arrow, and downward arrow. The arrows in the middle of the table from top to bottom include downward arrow, upward arrow, and downward arrow.

The reactions in the “Left section” from top to bottom are as follows:

$\text{IE}(\text{Na})$

$\text{E}(\frac{1}{2}\text{Cl}_2)$

$\Delta H_{\text{atom}}^\ominus(\text{Na})$

$\Delta H_f^\ominus(\text{NaCl})$

The reactions in the “Right section” from top to bottom are as follows:

$\text{EA}(\text{Cl})$

$\Delta H_{\text{lattice}}^\ominus$

The “Click” button on the bottom left allows users to check their answers. The number of correct answer(s) is indicated with a slider. When wrong answer(s) are input, the user can use the “Retry” button and redo the activity.

By following the directions of the arrows, we can see that there are two pathways to get to the step with the separated gaseous $\text{Na}^+(\text{g})$ and $\text{Cl}^-(\text{g})$.

Starting at the step $\text{Na}(\text{s}) + \frac{1}{2}\text{Cl}_2(\text{g})$, the first pathway is equal to the enthalpy change of formation plus the lattice enthalpy ($\Delta H_f^\ominus + \Delta H_{\text{lattice}}^\ominus$).

The second pathway is equal to the enthalpy of atomisation, plus the bond dissociation energy, plus the first ionisation energy, plus the electron affinity ($\Delta H_{\text{atom}}^\ominus + E + \text{IE} + \text{EA}$). This can be represented in equation form as:

$$\Delta H_f^\ominus + \Delta H_{\text{lattice}}^\ominus = \Delta H_{\text{atom}}^\ominus + E + \text{IE} + \text{EA}$$

According to Hess’s law, the enthalpy change for the two pathways is the same. By rearranging the equation, the lattice enthalpy can be calculated as shown:

$$\Delta H_{\text{lattice}}^\ominus = \Delta H_{\text{atom}}^\ominus + E + \text{IE} + \text{EA} - \Delta H_f^\ominus$$

Applied to the Born–Haber cycle in **Interactive 1**:

$$\Delta H_{\text{lattice}}^\ominus(\text{NaCl}) = \Delta H_{\text{atom}}^\ominus(\text{Na}) + \frac{1}{2}E(\text{Cl}_2) + \text{IE}(\text{Na}) + \text{EA}(\text{Cl}) - \Delta H_f^\ominus(\text{NaCl})$$

$$\Delta H_{\text{lattice}}^\ominus(\text{NaCl}) = +110 + 121 + 496 + (-349) - (-411)$$

$$\Delta H_{\text{lattice}}^\ominus(\text{NaCl}) = +789 \text{ kJ mol}^{-1}$$

Note that in this Born-Haber cycle, the bond enthalpy value E for Cl_2 is halved – this is because the formation of one mole of NaCl only requires one mole of chlorine atoms.

Worked example 1

Use the Born-Haber cycle to calculate the lattice enthalpy for potassium bromide.

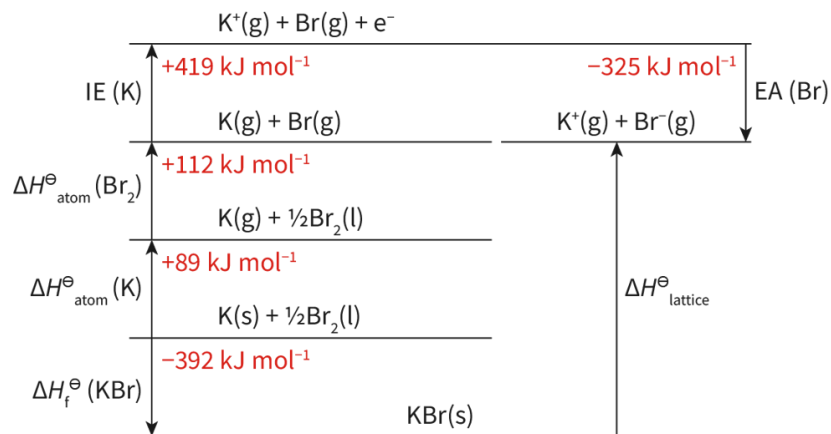


Figure 3. Born-Haber cycle for potassium bromide.

More information for figure 3

The image is a diagram representing the Born-Haber cycle for potassium bromide (KBr). It depicts a series of horizontal lines with arrows indicating energy changes during the formation of KBr. Starting from the bottom:

1. $\text{K(s)} + \frac{1}{2}\text{Br}_2(\text{l})$ with an energy change of $\Delta H_f^\ominus(\text{KBr}) = -392 \text{ kJ mol}^{-1}$, moving to $\text{K(g)} + \frac{1}{2}\text{Br}_2(\text{l})$ with $\Delta H^\ominus_{\text{atom}}(\text{K}) = +89 \text{ kJ mol}^{-1}$.
2. Moving upwards, $\text{K(g)} + \frac{1}{2}\text{Br}_2(\text{l})$ becomes $\text{K(g)} + \text{Br(g)}$ with an energy change $+112 \text{ kJ mol}^{-1}$.
3. This progresses to $\text{K}^+(\text{g}) + \text{Br(g)}$ with an ionization energy change of $\text{IE}(\text{K}) = +419 \text{ kJ mol}^{-1}$.
4. Finally, it transitions to $\text{K}^+(\text{g}) + \text{Br(g)} + \text{e}^-$ and an electron affinity change of $\text{EA}(\text{Br}) = -325 \text{ kJ mol}^{-1}$.
5. The final step shows the formation of KBr(s) , emphasizing the lattice energy $\Delta H^\ominus_{\text{lattice}}$.

The diagram includes the chemical symbols, different energy changes (in kJ/mol), and states of matter (solid, liquid, gas).

[Generated by AI]

$$\Delta H^\ominus_{\text{lattice}} = \Delta H^\ominus_{\text{atom}}(\text{K}) + \Delta H^\ominus_{\text{atom}}(\text{Br}_2) + \text{IE} + \text{EA} - \Delta H_f^\ominus$$

Therefore:

$$\begin{aligned}\Delta H^\ominus_{\text{lattice}} &= 89 + 112 + 419 - 325 - (-392) \\ &= 687 \text{ kJ mol}^{-1}\end{aligned}$$

Worked example 2

Use the Born-Haber cycle to calculate the enthalpy of formation for calcium oxide.

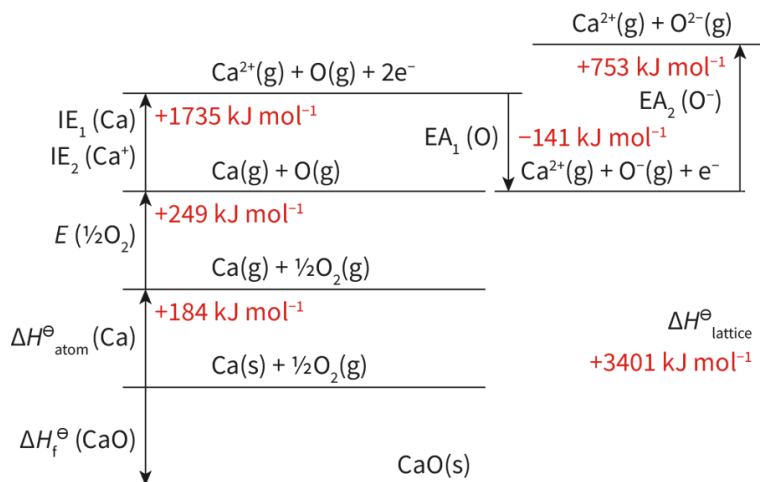


Figure 4. Born-Haber cycle for calcium oxide.

 More information for figure 4

The image depicts a Born-Haber cycle diagram for the formation of calcium oxide (CaO). It illustrates various enthalpy changes, each accompanied by their respective equations and energy values in kilojoules per mole (kJ/mol). The cycle transitions through several stages: starting from the formation of gaseous calcium ions (Ca^{2+}) and gaseous oxygen ions (O^{2-}) at the top, moving through intermediate stages that involve ionization energies (IE), electron affinities (EA), and other enthalpies (such as atomization and lattice enthalpies). The cycle is presented in a series of horizontal and vertical lines, each level representing a specific chemical reaction or physical change, labeled with the corresponding enthalpy, such as +1735 kJ/mol for the ionization energy of Ca and +3401 kJ/mol for the lattice enthalpy of CaO. The diagram visually and textually explains the stepwise formation of CaO by illustrating the various energy changes involved.

[Generated by AI]

$$\Delta H_{\text{lattice}}^{\ominus} = \Delta H_{\text{atom}}^{\ominus} + E + \text{IE} + \text{EA} - \Delta H_{\text{f}}^{\ominus}$$

Therefore:

$$\begin{aligned} \Delta H_{\text{f}}^{\ominus} &= \Delta H_{\text{atom}}^{\ominus} + E + \text{IE} + \text{EA} - \Delta H_{\text{lattice}}^{\ominus} \\ &= 184 + 249 + 590 + 1145 + 753 - 141 - 3401 \\ &= -621 \text{ kJ mol}^{-1} \end{aligned}$$

Worked example 3

Use the Born-Haber cycle to calculate the enthalpy of formation, ΔH_{f} , for sodium oxide.

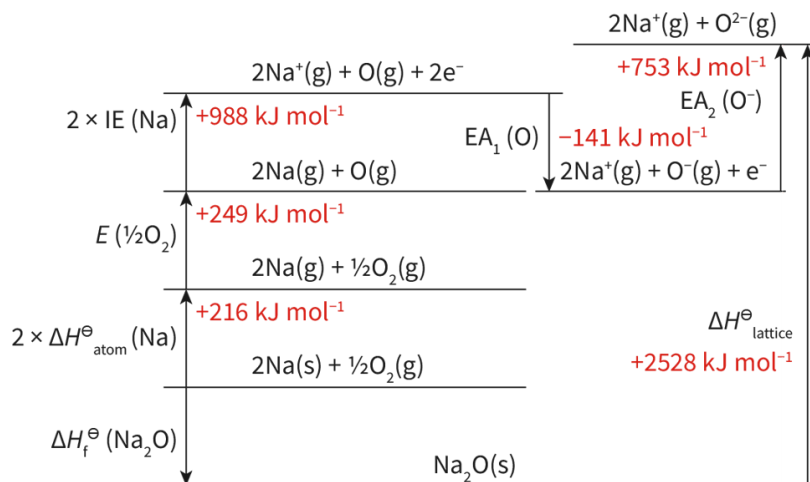


Figure 5. Born-Haber cycle for sodium oxide

 More information for figure 5

The Born-Haber cycle diagram illustrates the series of steps involved in the formation of sodium oxide (Na_2O) from its elements. It includes the following stages:

1. The process starts with 2 moles of solid sodium (Na) and half a mole of oxygen gas (O_2), which is represented as $(2\text{Na(s)} + \frac{1}{2}\text{O}_{2\text{(g)}})$ transitioning to 2 moles of gaseous sodium atoms $(2\text{Na(g)} + \frac{1}{2}\text{O}_{2\text{(g)}})$, with an enthalpy change of $+216 \text{ kJ mol}^{-1}$ labeled as $(2 \times \Delta H^\ominus_{\text{atom}}(\text{Na}))$.
2. This is followed by the excitation to sodium ions and oxygen atoms $(2\text{Na}^+\text{(g)} + \text{O(g)} + 2\text{e}^-)$ with an ionization energy of $+988 \text{ kJ mol}^{-1}$ labeled as $(2 \times \text{IE}(\text{Na}))$.
3. Next, the process involves the conversion of oxygen atoms to form negative ions $(2\text{Na}^+\text{(g)} + \text{O}^-\text{(g)} + \text{e}^-)$ with an electron affinity of -141 kJ mol^{-1} labeled as $(\text{EA}_1(\text{O}))$.
4. This is followed by further electron gain $(2\text{Na}^+\text{(g)} + \text{O}^{2-}\text{(g)})$ with a second electron affinity of $+753 \text{ kJ mol}^{-1}$ labeled as $(\text{EA}_2(\text{O}^-))$.
5. The final step is the formation of the ionic lattice $(\Delta H^\ominus_{\text{lattice}} = +2528 \text{ kJ mol}^{-1})$.

The overall enthalpy of formation ($\Delta H^\ominus_f(\text{Na}_2\text{O})$) is derived by summing these steps together to yield the enthalpy change for the process.

[Generated by AI]

$$\Delta H^\ominus_{\text{lattice}} = \Delta H^\ominus_{\text{atom}} + E + \text{IE} + \text{EA} - \Delta H^\ominus_f$$

Therefore:

$$\begin{aligned} \Delta H^\ominus_f &= \Delta H^\ominus_{\text{atom}} + E + \text{IE} + \text{EA} - \Delta H^\ominus_{\text{lattice}} \\ &= 216 + 249 + 988 + 753 - 141 - 2528 \\ &= -463 \text{ kJ mol}^{-1} \end{aligned}$$

Factors that affect the lattice enthalpy of an ionic compound

The magnitude of the lattice enthalpy of an ionic compound depends on two factors; the charge on the ions and the radius of the ions. The greater the charge and the smaller the radius, the higher the value of the lattice enthalpy. The table below has data for two ionic compounds, NaCl and KCl.

Compound	Cation radius ($\times 10^{-12}$ m)	Anion radius ($\times 10^{-12}$ m)	$\Delta H^\ominus_{\text{lattice}}$
NaCl	102	181	790
KCl	138	181	720

From the table, we can see that the difference in the lattice enthalpy is a result of the difference in the cation radius. The sodium ion, Na^+ , has a smaller radius than the potassium ion, K^+ . This results in a stronger attraction between the ions and a higher lattice enthalpy value.

With regards to the effect of ionic charge, we can compare the lattice enthalpies of NaCl and CaCl_2 , which are 790 kJ mol^{-1} and 2271 kJ mol^{-1} respectively. The higher lattice enthalpy of CaCl_2 is a result of the $2+$ charge of the calcium ion, Ca^{2+} , compared to the $1+$ charge of the sodium ion, Na^+ .

Worked example 4

State and explain which ionic compound has the highest lattice enthalpy, LiF or RbF.

Li^+ has a smaller ionic radius than Rb^+ so LiF would have the highest lattice enthalpy.

Activity

- **IB learner profile attribute:** Knowledgeable
- **Approaches to learning:** Communication skills — Presenting data appropriately
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual/pair activity

Sometimes it is easier to retain knowledge when you are able to manipulate objects. For this activity, obtain a piece of cardstock or cardboard and write out the 'puzzle pieces' of the Born-Haber cycle.

- Use these puzzle pieces to create the Born-Haber cycles for the following compounds: LiCl, NaBr, MgO

- Use a pencil to write the equations for each part of the Born-Haber cycle on the cardboard.
- Shuffle the pieces and then rearrange them into the correct Born-Haber cycle. Challenge yourself by doing this without looking at your notes!
- Take a photo of each when you are done so that you can use the information for calculating the lattice enthalpy.